GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS
PHASE 3: Project Design Phase

Project theme: historical Olympic participation and medal performance.

Objective

Design the proposed solution, data flow, analytical outputs, and technology choices.

Problem–solution fit

The solution addresses the difficulty of comparing historical Olympic performance by transforming detailed athlete-event rows into consistent summary tables, visualizations, and documented insights.

Proposed workflow

CSV input → schema and missing-value checks → data cleaning → grouped summaries → charts and comparisons → validation → findings and documentation.

High-level data flow

- Input: Olympic athlete-event dataset.
- Processing: validate columns, inspect missing values, normalize categories where appropriate, and group records by year, season, NOC/team, sport, and event.
- Analysis: count athlete-event entries and medal records separately; avoid treating blank medal values as medals.
- Output: summary CSV files, visualizations, findings, and project documentation.

Technology stack

- Python for data processing and analysis.
- Pandas and NumPy for cleaning, grouping, and calculations.
- Matplotlib or Seaborn for visualizations.
- Jupyter Notebook or Google Colab for reproducible work.
- Excel or Power BI as optional tools for reviewing summaries and dashboards.

Design decisions

- Keep raw input separate from cleaned outputs.
- Use NOC codes for consistent country/team comparisons while noting historical naming changes.
- Report medal counts and participation counts as distinct measures.
- Check duplicate athlete-event records and missing values before interpreting results.
- Use charts that match the question, such as line charts for trends and bar charts for comparisons.

Key outputs

A documented processing pipeline, summary tables by year/NOC/sport, selected charts, and a clear record of assumptions.

Expected phase deliverable

Solution design, data flow, and technology choices.